

NAOMICHI YUTANI

Department of Planetology, Graduate School of Science, Kobe University, 1-1 Rokkodai-cho,
Nada-ku, Kobe, Hyogo 657-8501, Japan

yutaninm@gmail.com

yutaninm@people.kobe-u.ac.jp

EDUCATION AND EMPLOYMENT STATUS

Kagoshima University, Bachelor of Science, Astronomy and Physics Mar. & 2020

Kagoshima University, Master of Science, Astronomy Mar. & 2022

Thesis topic: The origin and evolution of Merger driven Dust Obscured Galaxies

Advisor: Prof., Director, Amanogawa Galaxy Astronomy Research Center, Keiichi Wada

Kagoshima University, Ph.D. in Science, Astronomy Mar. & 2025

Title: A Study on the Emergence of Active Galactic Nuclei with Diverse Host Galaxies

Advisor: Prof., Director, Amanogawa Galaxy Astronomy Research Center, Keiichi Wada

Kobe University, JSPS Research Fellowship (PD) Apr. & 2025 - Mar. & 2026

FUNDING

Japan Society for the Promotion of Science (JSPS): Apr. & 2024 - Mar. & 2026

JSPS Research Fellowship for Young Scientists “DC2” (1,500,000 JPY)

RESEARCH INTERESTS

Active Galactic Nuclei

Galaxy Co-Evolution

Galaxy Merger

Spiral Galaxy

HONORS & AWARDS

JSPS Research Fellowship for Young Scientists Apr. & 2024 – Mar. & 2026

TEACHING & ADVISING

Research Assistant, Kagoshima University Aug. & 2024 – Jan. & 2025

Research Assistant, Kagoshima University Sep. & 2022 – Jan. & 2023

PUBLICATIONS

1. **Dynamical Mechanism of Episodic Gas Accretion to the Central Region of Spiral Galaxies** Sep. & 2025
Naomichi Yutani, Nozomu Kawakatu, Takayuki Saitoh, Keiichi Wada
[10.48550/arXiv.2509.14487](https://arxiv.org/abs/10.48550/arXiv.2509.14487)
2. **Dust-obscured Galaxies with Broken Power-law Spectral Energy Distributions Discovered by UNIONS** Apr. & 2025
Taketo Yoshida, Tohru Nagao, Yoshiki Toba, Akatoki Noboriguchi, Kohei Ichikawa, Hendrik Hildebrandt, *Naomichi Yutani* et al.
[10.3847/1538-4357/add930](https://arxiv.org/abs/10.3847/1538-4357/add930)
3. **Apparent Effect of Dust Extinction on the Observed Outflow Velocity of Ionized Gas in Galaxy Mergers** Jan. & 2024
Naomichi Yutani, Yoshiki Toba, Keiichi Wada
[10.3847/1538-4357/ad0dfc](https://arxiv.org/abs/10.3847/1538-4357/ad0dfc)
4. **Origin and Evolution of Dust-obscured Galaxies in Galaxy Mergers** Sep. & 2022
Naomichi Yutani, Yoshiki Toba, Shunsuke Baba, Keiichi Wada
[10.3847/1538-4357/ac87a2](https://arxiv.org/abs/10.3847/1538-4357/ac87a2)
5. **The eROSITA Final Equatorial-Depth Survey (eFEDS). A multiwavelength view of WISE mid-infrared galaxies/active galactic nuclei** May & 2022
Yoshiki Toba, Teng Liu, Tanya Urrutia, Mara Salvato, Junyao Li, Yoshihiro Ueda, Marcella Brusa, *Naomichi Yutani* et al.
[10.1051/0004-6361/202141547](https://arxiv.org/abs/10.1051/0004-6361/202141547)
6. **Mass Accretion toward Black Holes in the Final Phase of Galaxy Mergers** Feb. & 2020
Takeru Kawaguchi, *Naomichi Yutani*, Keiichi Wada
[10.3847/1538-4357/ab655a](https://arxiv.org/abs/10.3847/1538-4357/ab655a)

SEMINARS

1. Invited Speaker, “Dynamical Mechanism of Episodic Gas Accretion to the Central Region of Spiral Galaxies”, CPS seminar at CPS Conference Room and Online, 30 June 2025
2. Invited Speaker, “Mass Transport Mechanism from the Galactic Scale to the Galactic Nucleus Revealed by High Mass Resolution Calculations”, Seminar at Ehime University, 2 July 2024

INTERNATIONAL CONFERENCE

1. Invited Speaker, "Self-Regulated Convergence to the Local SMBH–Bulge Relation by Galaxy Mergers", Aug. & 2025
Naomichi Yutani, Takayuki Saitoh,
11th GALAXY EVOLUTION WORKSHOP, Nagoya University
2. "The Origin Of Molecular Clouds in the CND", Mar. & 2025
Naomichi Yutani, Nozomu Kawakatu, Takayuki Saitoh, Keiichi Wada,
Behind The Curtain Of Dust V (BCD-V), Ice Hotel (Jukkasjarvi, Sweden)
3. "Apparent Effect of Dust Extinction on the Observed Outflow Sep. & 2023
Velocity of Ionized Gas in Galaxy Mergers",
Naomichi Yutani, Yoshiki Toba, Keiichi Wada,
East-Asia AGN Workshop 2023, Kagoshima University
4. "IR pseudo-observations of merger driven DOGs and their evolution", Oct. & 2021
Naomichi Yutani, Keiichi Wada, Yoshiki Toba,
East-Asia AGN Workshop 2021, China (Online)
5. "IR pseudo-observations of merger driven DOGs and their evolution", Feb. & 2021
Naomichi Yutani, Keiichi Wada, Yoshiki Toba,
Galaxy Evolution Workshop 2020, Kagoshima University
6. "IR pseudo-observations of merger driven DOGs and their evolution", Jan. & 2021
Naomichi Yutani, Keiichi Wada, Yoshiki Toba,
CON-quest workshop, Sweden (Online)

DOMESTIC CONFERENCE

1. "Mass Supply to the Galactic Center by Spiral Arms: Mar. & 2025
On the Origin of Molecular Gas in Galactic Centers",
Naomichi Yutani, Keiichi Wada, Nozomu Kawakatu, Takayuki Saitoh,
Japanese Astronomical Society Spring Meeting 2025, Mito Citizen Hall
2. "Mass Supply to the Galactic Center by Spiral Arms: Dec. & 2024
On the Origin of Molecular Gas in Galactic Centers",
Naomichi Yutani,
37th Theoretical Symposium "Cosmology 100 years: Past and Future", National
Astronomical Observatory of Japan, Mitaka Campus
3. "Spiral and Cloud Formation in the Disk and Angular momentum Nov. & 2024
transport to the Galactic Center",
Naomichi Yutani,

- GMC Lifetime Mini Workshop, Kagoshima University
4. "Mass Supply Process by Spiral arms and Gas clumps in Isolated galaxy disks and the Emergence of Active Galactic Nuclei",
Naomichi Yutani,
ALMA Workshop 2024: AGN Feeding and Feedback in Massive Galaxies at the Centers of Galaxy Clusters, Kagoshima University Nov. & 2024
 5. "Mass Transport Mechanism from the Galactic Scale to the Galactic Nucleus Revealed by High Mass Resolution Calculations",
Naomichi Yutani, Nozomu Kawakatu, Takayuki Saitoh, Keiichi Wada,
Japanese Astronomical Society Autumn Meeting 2024, Kansai Gakuin University Sep. & 2024
 6. "Classification of Explosive Star Formation and Obscured Active Galactic Nuclei using Infrared Continuum",
Naomichi Yutani, Takayuki Saitoh, Satoshi Yamada, Shunsuke Baba, Keiichi Wada
Japanese Astronomical Society Spring Meeting 2024, University of Tokyo (Online) Mar. & 2024
 7. "Explosive Star Formation and Classification of buried Active Galactic Nuclei Using Infrared Continuum Radiation",
Naomichi Yutani,
Ehime-Kumamoto-Kagoshima Joint Research Presentation, Kagoshima University Feb. & 2024
 8. "Nature of Photoionized Outflow in late stages, of gas-rich Galaxy Mergers"
Naomichi Yutani, Yoshiki Toba, Keiichi Wada
Japanese Astronomical Society Autumn Meeting 2023, Nagoya University Sep. & 2023
 9. "Evolution of Photoionized Outflow velocity in the late stages of Galaxy Collisions",
Naomichi Yutani, Yoshiki Toba, Keiichi Wada
Japanese Astronomical Society Spring Meeting 2023, Rikkyo University Mar. & 2023
 10. "Origin and Evolution of Dust-obscured Galaxies in Galaxy Mergers",
Naomichi Yutani, Yoshiki Toba, Shunsuke Baba, Keiichi Wada,
Japanese Astronomical Society Autumn Meeting 2022, Niigata University Sep. & 2022
 11. "Origin and Evolution of Dust-obscured Galaxies",
Naomichi Yutani, Yoshiki Toba, Shunsuke Baba, Keiichi Wada
AGN mini-workshop, Kagoshima University Aug. & 2022
 12. "Origin and Evolution of Merger-driven Dust-obscured Galaxies",
Naomichi Yutani, Keiichi Wada, Yoshiki Toba,
Japanese Astronomical Society Spring Meeting 2021, Tokyo Institute of Technology Mar. & 2021
 13. "Infrared pseudo-observation of Merger-driven DOGs Jan. & 2021

considering AGN-feedback",

Naomichi Yutani, Keiichi Wada, Yoshiki Toba,

CfCA Users Meeting 2021, Tokyo Institute of Technology

POSTERS

1. "Mass Transport Mechanism to Galactic Nuclei in Spiral Galaxies: June & 2024
Effect of Supernova Explosions",
Naomichi Yutani, Keiichi Wada,
Mini Workshop On Star Formation, Kagoshima University
2. "The Origin and Evolution of Dust Obscured Galaxies Dec. & 2023
in the late stages of galaxy mergers",
Naomichi Yutani, Yoshiki Toba, Shunsuke Baba, Keiichi Wada,
36th Theoretical Symposium, Hirosaki University
3. "Origin and Evolution of Dust-obscured Galaxies in Galaxy Mergers", Dec. & 2022
Naomichi Yutani, Yoshiki Toba, Shunsuke Baba, Keiichi Wada,
HSC-AGN f2f meeting, Kagoshima University